✅ 1. Competency #1: Select a mathematical description or representation that describes a problem

- Highest relevance.
- Specific Analysis:
  - This is the core objective of the task. Given a real-world business rule ("selling price includes a 37.5% profit on wholesale cost"), students are asked to select the correct algebraic formula that represents this rule.
  - Students must mentally construct the correct model: $s = w + 37.5\% \times w$.
  - The subsequent judgment process (the task) is essentially validating whether the given mathematical descriptions (formulas) are consistent with the constructed model. This directly corresponds to the *formulate.pdf* competency of "selecting an appropriate mathematical description for a problem."

✅ 2. Competency #2: Identify the key variables in a model

- Highly relevant.
- Specific Analysis:
  - While the variables w and s are provided, students must understand their real-world significance within the model:
    - w (Wholesale Price) is the base (100%).
    - s (Selling Price) is the result (137.5%).
  - Error Trap Analysis:
    - The first incorrect option, $s = w + 0.375$, confuses "percentage" with "absolute amount." Students must recognize that 0.375 is a coefficient that must be multiplied by the base w , rather than a fixed amount of 0.375 zeds.
  - This discrimination of variable nature (proportion vs. absolute value) is fundamental to building a correct mathematical model.

✅ 3. Competency #5: Recognize mathematical structure including regularities, relationships, and patterns

- Highly relevant.
- Specific Analysis:
  - The task examines the mathematical structure of percentage increase.
  - Students need to recognize that "adding 37.5%" is algebraically equivalent to "multiplying by 1.375" (i.e., Formula 3: $s = 1.375w$).
  - Furthermore, students must recognize the structure of the inverse operation: If s is 137.5% of w , then w must be $s/1.375$ (approximately 0.727 s), not simply $1 - 0.375 = 0.625$ (i.e., Formula 4 is incorrect; it falsely assumes symmetry in percentage change).
  - Recognizing this "multiplicative relationship" versus an "additive relationship," and understanding inverse operations, is key to identifying mathematical structure.

✅ 4. Competency #6: Identify and describe the mathematical aspects of a real-world problem situation

- Highly relevant.
- Specific Analysis:
  - The problem description is in natural language. Students must distill it into pure mathematical facts:
    - "Normal selling price includes a profit of 37.5%" $\rightarrow s = w + 37.5\% \times w$.
    - "The profit is calculated as a percentage of the wholesale price" $\rightarrow$ Crucial Constraint: The base is w , not s .
  - The process of filtering out extraneous context (MP3s, headphones) and extracting the core mathematical logic (profit calculation based on cost) is exactly what *formulate.pdf* defines as "identifying the mathematical aspects of a real-world situation."

✅ 5. Competency #9: Translate a problem into a standard mathematical representation or algorithm

- Moderately relevant.
- Specific Analysis:
  - Although the task is multiple-choice rather than open-response, students are mentally performing translation when evaluating each formula.
  - For example, translating the sentence into $s = w(1 + 0.375)$.
  - Then, to check Option 2 ($w = s - 0.375s$), students need to manipulate the standard form or rearrange the option to verify equivalence. This "form conversion" and "equivalence checking" demonstrates the ability to translate a problem into standard mathematical representations.

✅ 1. 能力 #1：选择描述问题的数学描述或表示 (Select a mathematical description or representation that describes a problem)

- 最高相关性。
- 具体分析：
  - 这是该题目的核心任务。题目给出了现实世界的商业规则（“售价包含批发价37.5%的利润”），要求学生选择正确的代数公式来表示这一规则。
  - 学生必须在脑海中构建正确的模型： $s = w + 37.5\% \times w$ 。
  - 随后的判断过程（Task）本质上是验证给定的数学描述（公式）是否与选定的模型一致。这直接对应了 *formulate.pdf* 中“为问题选择适当的数学描述”的能力。

✅ 2. 能力 #2：识别模型中的关键变量 (Identify the key variables in a model)

- 高度相关。
- 具体分析：
  - 题目已经定义了 w 和 s ，但学生必须理解这两个符号背后的现实意义：
    - w (Wholesale Price) 是基数（100%）。
    - s (Selling Price) 是最终结果（137.5%）。
  - 错误陷阱分析：
    - 第一个错误选项 $s = w + 0.375$ 混淆了“百分比”和“具体金额”。学生必须识别出 0.375 是一个比例系数，必须乘以基数 w ，而不是一个固定的 0.375 zeds。
  - 这种对变量性质（比例 vs. 绝对值）的敏锐识别，是构建数学模型的基础。

✅ 3. 能力 #5：识别问题或情境中的数学结构（规律、关系、模式）（Recognise mathematical structure including regularities, relationships, and patterns)

- 高度相关。
- 具体分析：
  - 题目考察的是百分比增长的数学结构。
  - 学生需要识别出“加 37.5%”在代数上等同于“乘以 1.375”（即公式3 $s = 1.375w$ ）。
  - 同时，学生需要识别出逆运算的结构：如果 s 是 w 的 137.5%，那么 w 必须是 s 的 $1/1.375 \approx 0.727$ ，而不是简单的 $1 - 0.375 = 0.625$ （即公式4是错误的，它错误地假设了百分比在两个方向上是对称的）。
  - 这种对“乘法关系”而非“加法关系”的识别，以及对“反向关系”的理解，是识别数学结构的关键。

✅ 4. 能力 #6：识别现实问题中的数学方面，包括显著变量 (Identify and describe the mathematical aspects of a real-world problem situation)

- 高度相关。
- 具体分析：
  - 题目中的文字描述是自然语言，学生必须将其剥离为纯粹的数学事实：
    - “Normal selling price includes a profit of 37.5%” $\rightarrow s = w + 37.5\% \times w$ 。
    - “The profit is calculated as a percentage of the wholesale price” $\rightarrow$ 关键约束：基准是 w ，而不是 s 。
  - 这种将冗余的商业背景（MP3、耳机等）过滤掉，提取出核心数学逻辑（基于成本的利润率计算）的过程，正是 *formulate.pdf* 中关于“识别现实问题中的数学方面”的定义。

✅ 5. 能力 #9：将问题转化为标准的数学表示或算法 (Translate a problem into a standard mathematical representation or algorithm)

- 中等相关性。
- 具体分析：
  - 虽然题目是判断题而非填空题，但学生在判断每一个公式时，实际上是在脑海中进行转化。
  - 例如，将自然语言句子转化为 $s = w(1 + 0.375)$ 。
  - 然后，学生需要将这个标准形式与选项2 $w = s - 0.375s$ 进行对比，这需要将标准形式变形，或者将选项2变形，看是否等价。这种“形式转换”和“等价性验证”体现了将问题转化为标准数学表示的能力。